INCH - POUND
MIL-V-47308A (MI)
18 October 1991
SUPERSEDING
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8 July 1987

MILITARY SPECIFICATION

VALVE, SERVO, HYDRAULIC FLOW CONTROL

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 scope. This specification describes the requirements for one type of servo hydraulic flow control valve.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC *N/A* FSC 4810
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2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2) .

SPECIFICATIONS

MILITARY

MIL-V-3	-	Valves, Fittings, and Flanges (Except for Systems Indicated Herein); Packaging of
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base: Aircraft, Missile, and Ordnance
MIL-H-8775	-	Hydraulic System Components, Aircraft and Missiles, General Specification for

STANDARDS

MILITARY

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-454	-	Standard General Requirements for Electronic Equipment
MIL-STD-810	-	Environmental Test Methods and Engineering Guidelines
MS3116	-	Connectors, Plug, Electric, Series 1, Solder Type, Straight, Bayonet Coupling
MS28775	-	Packing Preformed, Hydraulic 135°C (“O” Ring)

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Materials. Materials to be used in the manufacture of the valve shall be selected in accordance with the requirements of MIL-H-8775 for type II components.

3.3 Design. Design of the valve shall conform to figure 1 (see 4.6.1) .

3.3.1 Valve action. When voltage potential at pin A is positive with respect to the potential at pin C and coils are series connected as shown on the wiring diagram of figure 1, fluid flow shall be out of port 1 (see 4.6.1).

3.3.2 Fluid media. The valve shall be designed to operate with hydraulic fluid conforming to MIL-H-5606 (see 4.6.1).

3.4 Construction. Construction of the valve shall conform to MIL-H-8775 (see 4.6.1).

3.5 Performance.

3.5.1 Insulation resistance and dielectric strength. The insulation resistance between the coils and valve body shall be greater than 50 megohms following a 15 second application of 1000 volts direct current (Vdc) (see 4.6.2).

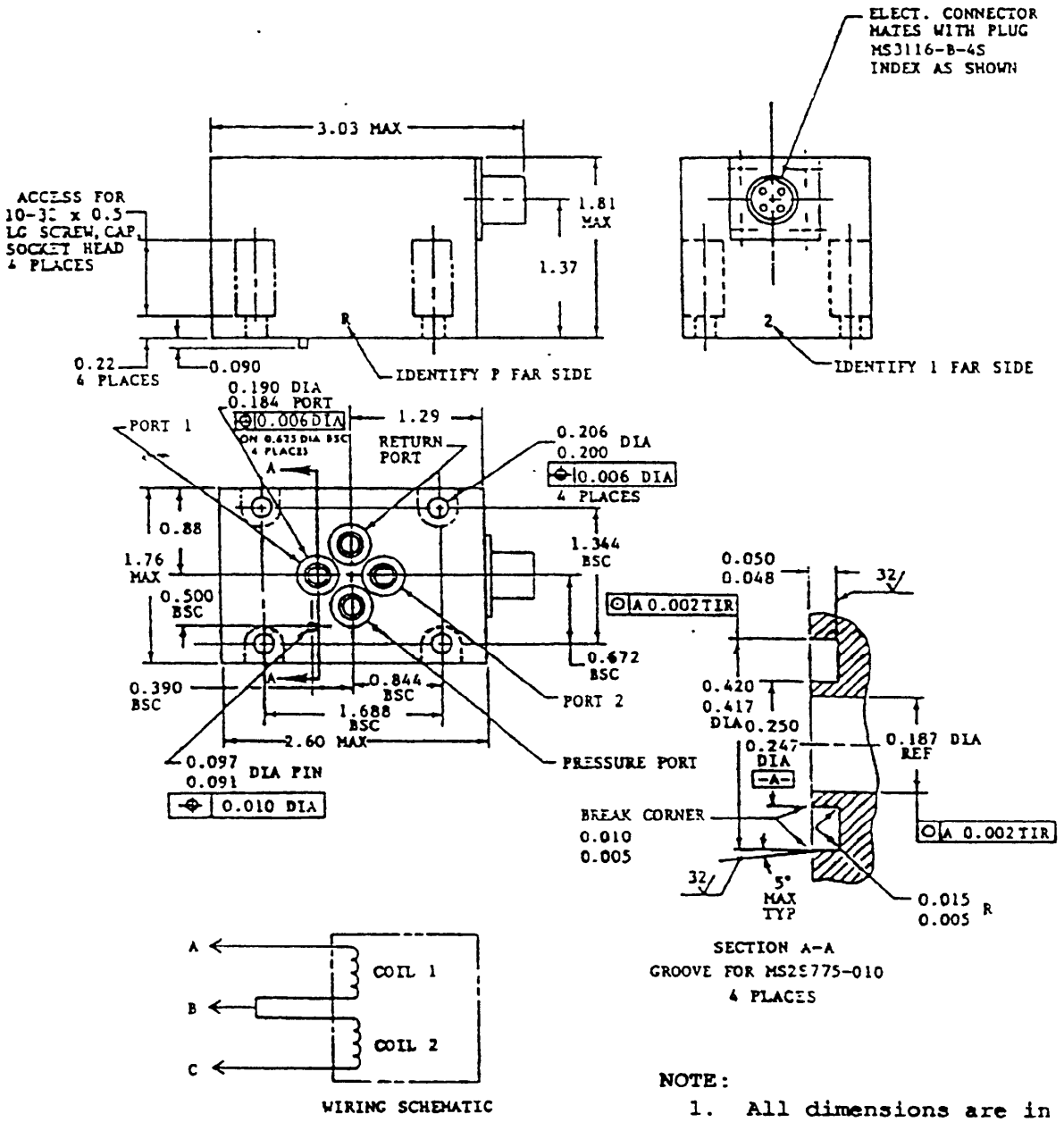
3.5.2 Coil resistance. The direct current (de) resistance of each coil shall be 1000 $\pm$ 100 ohms (see 4-6.3).

3.5.3 Coil apparent inductance. The greatest apparent inductance of the two coils series connected shall be less than or equal to 12 henries (H) with a coil current of one milliampere (ma) root mean square (rms) at 150 hertz (Hz) (see 4.6.4).

3.5.4 Proof pressure. The valve shall withstand without external leakage or permanent performance degradation, 1500 pounds force per square inch (lbf/in^2) applied to the pressure port, port 1 and port 2 with the return port open for 2 minutes, followed by 1000 lbf/in^2 applied simultaneously to all ports for 2 minutes (see 4.6.5).

3.5.5 External leakage. There shall be no evidence of external leakage, other than a slight wetting insufficient to form a drop at an operating pressure of 1000 lbf/in^2 (see 4.6.6).

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NOTE:

1. All dimensions are in inches.

FIGURE 1. Servo valve, hydraulic flow control.

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3.5.6 Internal leakage. The internal leakage of the valve shall be not *more* than 0.15 gallons per minute (gpm) at 1000 lbf/in^2 and any applied control current (see 4.6.7).

3.5.7 Rated flow. The rated flow shall be ± 1.75 gpm at no-load conditions with a valve pressure drop and ± 4 ma coil current through series adding coils (see 4.6.8 and 4.7.1 through 4.7.5).

3.5.8 Normal flow gain. The normal flow gain with coil series connected shall be 0.437 ± 0.04 gpm per ma (see 4.6.8).

3.5.9 Linearity. Deviation of the normal flow from the normal flow gain shall be not greater than 0.4 ma (see 4.6.8).

3.5.10 Hysteresis. The difference in input currents required to produce the same valve output when the input is cycled for a period greater than or equal to 15 seconds shall be not greater than 0.16 ma (see 4.6.8).

3.5.11 Null bias. Null bias control current shall be not greater than 0.12 ma (see 4.6.8).

3.5.12 Threshold. The maximum threshold shall be not greater than 0.02 ma (see 4.6.9).

3.5.13 Pressure gain. The blocked port pressure gain shall be not greater than 8000 lbf/in^2 differential (d) per ma as determined by the slope of the line connecting the points on the curve between ± 600 lbf/in^2 (d) (see 4.6.10).

3.5.14 Frequency response. With a ± 2 ma sinusoidal signal applied, the valve amplitude ratio and phase lag shall be within the limits shown on figure 2 (see 4.6.11 and 4.7.1 through 4.7.5).

3.5.15 Step response. The valve shall have a step response of not greater than 5 milliseconds (ins) . The step response shall be the tine required for the valve output to change from rated flow of one polarity to 90 percent of rated flow of the opposite polarity with an input of a ± 4 ma, 20 Hz square wave input (see 4.6.12).

3.5.16 Null shift with temperature variation. The null shift over the fluid temperature range of -31 to 135° Celsius (C) shall be not greater than ± 0.16 ma (see 4.6.13).

3.5.17 Null shift with life. The null bias shall be not greater than 0.16 ma at any time during the rated life of the valve (see 4.6.14).

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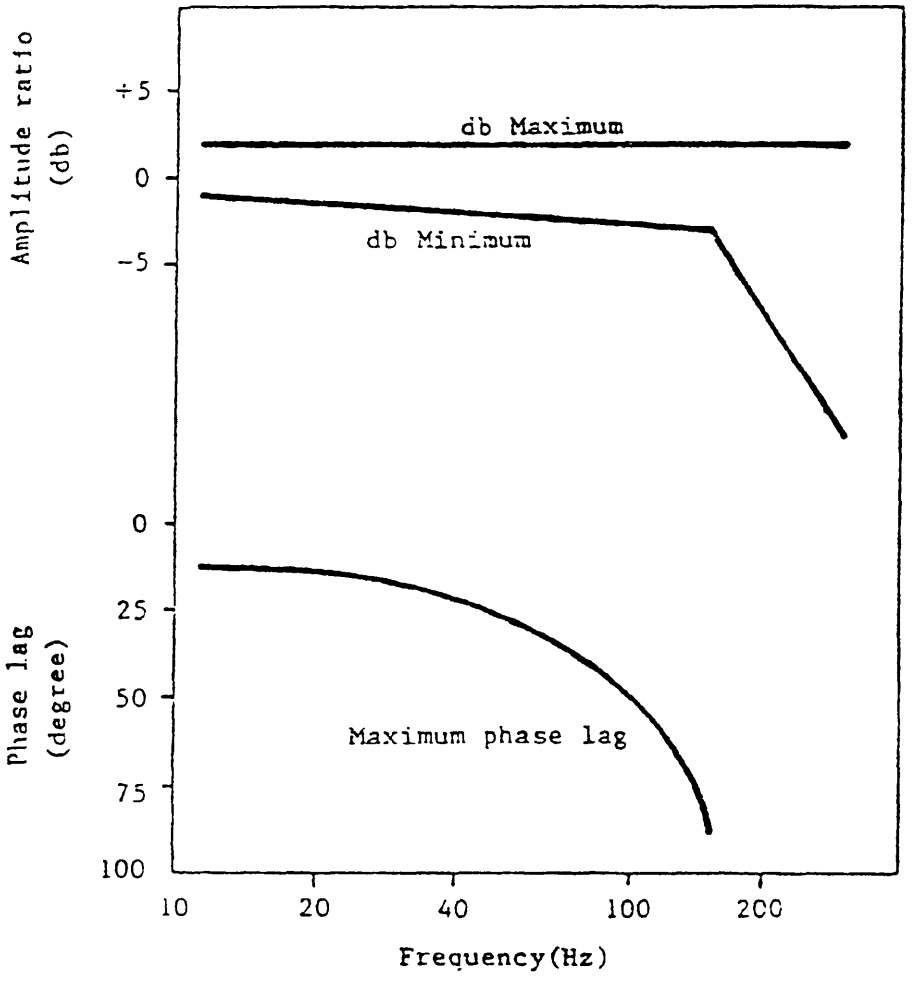


FIGURE 2. Frequency response.

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3.5.18 Null shift with supply pressure variation. The null shift with a variation of supply pressure of ± 50 lbf/in^2 shall be not greater than 0.02 ma (see 4.6.15).

3.5.19 Null shift with return pressure variation. The null shift with a variation of return pressure from O to 1000 lbf/in^2 shall be not greater than 0.04 ma (see 4.6.16).

3.6 Environmental service conditions. The valve shall operate satisfactorily, as specified, during or after exposure to the following environmental conditions (see 4.7) .

3.6.1 Temperature. The valve shall perform as specified while subjected to a temperature range of -31.7 to 5.17°C with a fluid temperature range of -31.7 to 135°C (see 4.7.1 and 4.7.2).

3.6.2 Vibration. The valve shall perform as specified while being subjected to sinusoidal vibration of 0.10 inch double amplitude (da) at 5 to 10 Hz, ± 0.51 gravity (g) at 10 to 13 Hz, 0.060 inch da at 13 to 55 Hz and ± 10 g at 55 to 500 Hz (see 4.7.3).

3.6.3 Shock. The valve shall perform as specified after being subjected to three 20 g, 11 ms half-sine shocks in each of the three orthogonal axes (see 4.7.4) .

3.6.4 Life. The valve shall have an operating life of not less than 2000 hours with an input signal of ± 1.0 ma sine wave, 10 Hz applied and an operating fluid temperature of 93.3°C (see 4.7.5) .

3.7 Burst pressure. The valve shall not rupture with a burst pressure of 4000 lbf/in^2 applied to port B, port 1 and 2 with the return port open and followed by 2000 lbf/in^2 applied to all ports. The valve shall not be required to operate properly following application of burst pressure (see 4.8).

3.8 Marking. The valve shall be marked in accordance with MIL-STD-130 (see 4.6.1 and Section 5).

3.9 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9 (see 4.6.1).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible fox the performance of all inspection requirements (examinations and tests) as specified herein.

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Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspection conditions shall be performed in accordance with the conditions of 4.3.1, 4.3.2, 4.3.3 and 4.3.4.

4.3.1 Temperature. Temperature shall be room ambient 26.7 ± 4°C.

4.3.2 Barometric pressure. Barometric pressure shall be facility ambient 28 to 32 inches of mercury (in Hg).

4.3.3 Humidity. Relative humidity shall be not greater than 95 percent.

4.3.4 Fluid. Fluid media shall be in accordance with 3.3.2.

4.4 First article inspection. First article inspection shall consist of the examination and tests as specified in table I and shall be conducted in the order specified in table II. Quantity production shall be withheld until the first article sample has been pronounced satisfactory by the procuring activity (see 3.1 and 6.3).

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4.4.1 First article sample. The first article sample shall consist of three units, representative of the production product. The first article samples shall be subjected to inspections and tests specified in table I with the order of inspection and sample allocation as shown in table II.

TABLE I. First article inspection.

Inspection	Requirement paragraph	Test method paragraph
Null shift with temperature variation	3.5.16	4.6.13
Null shift with life	3.5.17	4.6.14
Null shift with supply pressure variation	3.5.18	4.6.15
Null shift with return pressure variation	3.5.19	4.6.16
Low temperature	3.6.1	4.7.1
High temperature	3.6.1	4.7.2
Vibration	3.6.2	4.7.3
Shock	3.6.3	4.7.4
Life	3.6.4	4.7.5
Burst pressure	3.7	4.8

TABLE II. First article inspection sequence.

Sample	Inspection and Sequence
1 and 2	Low temperature High temperature Null shift with temperature variation Null shift with supply pressure variation Null shift with return pressure variation Vibration Shock
3	Life Null shift with life Burst pressure

4.5 Quality conformance inspection. Quality conformance inspection shall be conducted on the basis of single sampling of a random sample selected from each inspection lot or portion thereof (see 6.2).

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4.5.1 Inspection lot. The lot size shall be not greater than the quantity of items produced in one continuous operation utilizing the same production processes for delivery for one contract or order.

4.5.2 Inspection and tests. Samples selected in accordance with 4.5 shall be subjected to the inspections and tests in table III in that order.

TABLE III. Quality conformance inspection.

Inspection	Requirement paragraph	Test method paragraph
Examination of product	3.3, 3.4, 3.8 & 3.9	4.6.1
Insulation resistance/dielectric strength	3.5.1	4.6.2
Coil resistance	3.5.2	4.6.3
Coil apparent inductance	3.5.3	4.6.4
Proof pressure	3.5.4	4.6.5
External leakage	3.5.5	4.6.6
Internal leakage	3.5.6	4.6.7
Rated flow	3.5.7	4.6.8
Normal flow gain	3.5.8	4.6.8
Linearity	3.5.9	4.6.8
Hysteresis	3.5.10	4.6.8
Null bias	3.5.11	4.6.8
Threshold	3.5.12	4.6.9
Pressure gain	3.5.13	4.6.10
Frequency response	3.5.14	4.6.11
Step response	3.5.15	4.6.12

4.6 Method of inspection.

4.6.1 Examination of product. Each valve shall be visually and dimensionally inspected and polarity determined to assure conformance to 3.3, 3.4, 3.8 and 3.9.

4.6.2 Insulation resistance and dielectric strength. A voltage of 1000 Vdc shall be applied across coil 1 and coil 2 for 15 seconds. After 15 seconds, the resistance from point A, point B, or point C to the valve body (see figure 1) shall be measured for conformance to 3.5.1.

4.6.3 Coil resistance. The resistance across coil 1 and coil 2 shall be measured for conformance to 3.5.2.

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4.6.4 Coil apparent inductance. A 1.0 ma rms signal of 150 Hz shall be applied across coil 1 and coil 2 with the coils series connected and the inductance measured for conformance to 3.5.3.

4.6.5 Proof pressure. A *1500* **lbf/in^2** pressure shall be applied to the pressure port, port 1 and port 2 with the return port open for 2 minutes followed by 1000 **lbf/in^2** applied simultaneously to all ports for 2 minutes to determine conformance to 3.5.4.

4.6.6 External leakage. With 1000 **lbf/in^2** applied to the valve, external leakage shall be measured to determine conformance to 3.5.5.

4.6.7 Internal leakage: with the valve in a blocked output port configuration and 1000 **lbf/in^2** applied, the return flow versus input current through series connected coils shall be plotted over a range of -4 to 4 ma. The leakage shall be measured for conformance to 3.5.6.

4.6.8 Rated flow, normal flow gain, **lbf/in^2(d)** linearity, hysteresis and null bias. With a differential pressure of 980 **lbf/in^2(d)** established across the valve with a no load condition, the input current to the valve coils shall be cycled through ±4 ma in 15 seconds. A flow output versus current curve shall be plotted to determine conformance to 3.5.7 through 3.5.11.

4.6.9 Threshold. Using the valve in the configuration shown in 4.6.8, determine the lowest current change required to change the output flow at 1.0 ma intervals over the range of ± 4 ma to determine conformance to 3.5.12.

4.6.10 Pressure gain. With the valve in the blocked port configuration, the output differential pressure versus input current through the series connected coils shall be plotted over the input current range of ± 4.0 ma to determine conformance to 3.5.13.

4.6.11 Frequency response. A ± 2.0 ma sinusoidal input signal shall be applied through the series connected coils and a gain and phase frequency response shall be plotted from 10 to 200 Hz to determine conformance *to* 3.5.14.

4.6.12 Step response. Using the valve in the configuration shown in 4.6.8, apply a ± 4 ma, 20 Hz square wave input and measure the time required *for* the flow rate to change 90 percent of the flow rate of the opposite polarity to determine conformance to 3.5.15.

4.6.13 Null shift with temperature variation. The valve shall be subjected to temperature variation from -31.7° to 135° to -31.7°C and null bias measured every 11.1°C to determine conformance to 3.5.16.

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4.6.14 Null shift with life. During the life test (see 4.7.5), record null bias at *200* hour intervals and at the end of 2000 hours to determine conformance to *3.5.17.*

4.6.15 Null shift with supply pressure variation. A supply pressure of 1000 lbf/in^2 shall be applied and varied $\pm$ 50 lbf/in^2. The null bias shall be measured at each variation to determine conformance to 3.5.18.

4.6.16 Null shift with return pressure variation. The return pressure shall be varied from O to 0.689 and the null bias measured at 20 lbf/in^2 intervals to determine conformance to 3.5.19.

4.7 Environmental service conditions. The valve shall operate satisfactorily, as specified, during or after exposure to the requirements of 3.6.

4.7.1 Low temperature. The valve shall be tested in accordance with MIL-STD-810, method 502, procedure 1 and the requirements of 3.6.1 to determine conformance to 3.5.7 through 3.5.14.

4.7.2 High temperature. The valve shall be tested in accordance with MIL-STD-810, method 501, procedure 1 and the requirements of 3.6.1 to determine conformance to 3.5.7 through 3.5.14.

4.7.3 Vibration. The valve shall be tested in accordance with MIL-STD-810, method 514, category 5 and the requirements of 3.6.2 to determine conformance to 3.5.7 through 3.5.14.

4.7.4 Shock. The valve shall be tested in accordance with MIL-STD-810, method 516, procedure 1 and 3.6.3 to determine conformance to 3.5.7 through 3.5.14.

4.7.5 Life. The valve shall be tested in accordance with 3.6.4, at 200 hours and at the end of 2000 hours to determine conformance with 3.5.7 through 3.5.14.

4.8 Burst pressure. The valve shall be tested in accordance with 3.7. The valve shall not rupture during application of burst pressure; however, the valve shall not be required to operate properly after the application.

4.9 Inspection of packaging. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of MIL-V-3.

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5. PACKAGING

5.1 Preservation. Preservation shall be Level A *or C* in accordance with MIL-V-3.

5.2 Packaging and packing. Packaging and packing shall be Level A, B or C in accordance with MIL-V-3.

5.3 Marking. Marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve is intended for use as a flow control for a servo hydraulic actuator.

6.2 Acquisition requirements . Acquisition documents must specify the following:

a. Title, number, and date of the specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1)

c. Whether first article is required (see 3.1)

d. Sampling plan if other than specified (see 4.5)

e. Applicable level of preservation, packing and marking if other than specified (see section 5) .

6.3 First article. When first article inspection is required, the contracting officer should provide *specific* guidance to offerers whether the item(s) should be a first article sample, a first production item, or a number of items to be tested as specified *in* 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations *for* bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract (see 3.1).

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6.4 Metrication. Metric equivalents in accordance with FED-STD-376 are acceptable for use in this specification.

6.5 Subject term (keyword) listing.

Actuator
Fluid regulator
Pressure port

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army - MI

Preparing activity:
Army - MI

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INSTRUCTIONS

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-V-47308A (MI)	

3. DOCUMENT TITLE
VALVE, SERVO, HYDRAULIC FLOW CONTROL

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
COMMANDER U.S. ARMY MISSILE COMMAND	(1) Commercial (205) 876-6980	(2) AUTOVON 746-6980
c. ADDRESS (Include Zip Code) ATTN: AMSMI-RD-SE-TD-ST REDSTONE ARSENAL, AL 35898-5270	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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